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Programme:	AI & DS
Course Code & Course Name	CSA0904-Programming in Java
Academic Year / Batch	2023
Faculty Name	Dr.R.Thalapthi Rajasekaran
Assignment Title	Smart Library Management System
Date of Issue	01-09-2026
Date of Submission	02-09-2026
Maximum Marks	

1. Problem Statement and Problem Formulation

A library requires an efficient system to manage books, members, borrowing, reservations, overdue notifications, and report generation. Manual record maintenance is time-consuming and prone to errors such as duplicate entries, inaccurate inventory, and delayed notifications.

The proposed Smart Library Management System is a Java Applet-based GUI application that automates these activities using Object-Oriented Programming, Java Collections Framework, Exception Handling, and Multithreading. The system maintains book availability, supports reservations through a waiting list, generates reports, and sends due-date and overdue notifications while ensuring safe concurrent access to shared library resources.

2. Objective and Expected Outcomes

Objectives

- Develop a Java Applet-based GUI application.
- Apply Object-Oriented Programming concepts.
- Manage books and members efficiently.
- Automate book issue and return.
- Implement reservation and notification features.
- Generate circulation and inventory reports.
- Demonstrate multithreading and synchronization.

Expected Outcomes

- Accurate book inventory.
- Fast member registration.
- Automated fine calculation.
- Efficient reservation management.
- Reliable overdue notification.
- Reduced manual errors.

3. Requirements, Constraints and Assumptions

Functional Requirements

- Add books
- Register members
- Search books
- Issue books
- Return books
- Reserve books
- Cancel reservations
- Display books
- Display members
- Generate reports
- Send notifications

Constraints

- Java Applet GUI Application
- JDK 17
- CSV files used as persistent database
- Single administrator

Assumptions

- Book ID is unique.
- Member ID is unique.
- Fine is calculated only for overdue books.
- Notifications are displayed through the graphical user interface.

4. Application of Relevant Course Knowledge

The project demonstrates the following Java concepts:

- Classes and Objects
- Encapsulation
- Data Hiding
- Constructors
- Inheritance
- Polymorphism
- Method Overriding
- ArrayList
- HashMap
- HashSet
- Iterator
- Exception Handling
- User-defined Exceptions
- Multithreading
- Synchronization
- Generics

5. Design / Proposed Solution / Methodology

The system is divided into the following modules:

Book Management

- Add Book
- Search Book
- Display Books

Member Management

- Register Member
- Display Members

Transaction Module

- Issue Book
- Return Book

Reservation Module

- Reserve Book
- Cancel Reservation

Notification Module

- Due Notifications
- Overdue Notifications

Report Module

- Inventory Report
- Borrow Report
- Fine Report

6. Algorithm / Pseudocode

START

Display Main Menu

Repeat

Read User Choice

IF Choice = 1

Add Book

ELSE IF Choice = 2

Register Member

ELSE IF Choice = 3

Search Book

ELSE IF Choice = 4

Issue Book

ELSE IF Choice = 5

Return Book

ELSE IF Choice = 6

Reserve Book

ELSE IF Choice = 7

Cancel Reservation

ELSE IF Choice = 8

Display Books

ELSE IF Choice = 9

Display Members

ELSE IF Choice = 10

Generate Reports

ELSE IF Choice = 11

Send Notifications

ELSE IF Choice = 12

Exit

UNTIL Exit

STOP

7. Implementation / Environment / Tools Used

Software Used

Tool	Purpose
Java JDK 17	Programming Language and Runtime
IntelliJ IDEA	Code Development and Debugging
Antigravity AI	AI-assisted code generation and development support
Windows 11	Operating System

Development Environment

- Programming Language: Java
- JDK Version: 17
- IDE: IntelliJ IDEA
- Operating System: Windows 11
- Application Type: GUI-based Java Applet Application

8. Test Cases

Test Case	Expected Result	Actual Result	Status
Add Book	Book Added	Book Added	Pass
Register Member	Member Registered	Member Registered	Pass
Search Book	Book Found	Book Found	Pass
Issue Book	Book Issued	Book Issued	Pass
Return Book	Book Returned	Book Returned	Pass
Reserve Book	Reservation Created	Reservation Created	Pass
Generate Report	Report Displayed	Report Displayed	Pass

Notification	Message Displayed	Message Displayed	Pass

9. Execution Screenshots / Outputs

Figure 1: Main Menu

The Smart Library Management System starts by loading sample books, members, active loans, and reservation data. A menu is displayed with twelve options, allowing users to perform operations such as adding books, registering members, issuing and returning books, reserving books, generating reports, sending notifications, and exiting the application.

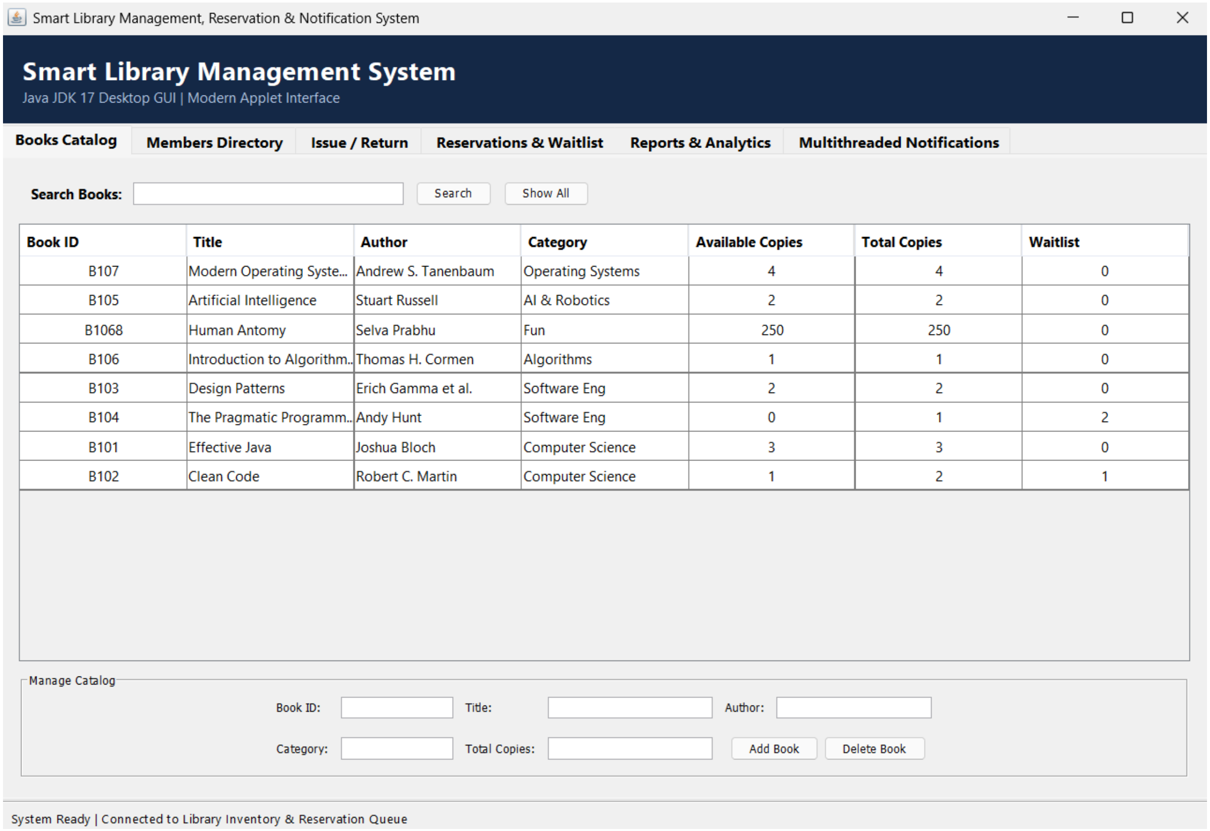


Figure 2: Add Book

The Add Book module allows the administrator to enter the details of a new book, including Book ID, title, author, category, and the number of available copies. After successful validation, the book is added to the library inventory, and a confirmation message is displayed.

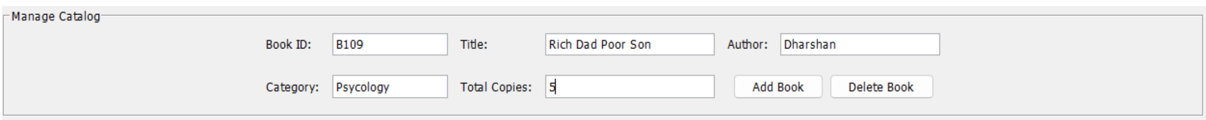


Figure 3: Register Member

The Register Member module enables the administrator to add a new library member by entering the member ID, name, contact details, and membership type. Once the information is validated, the member is successfully registered in the system.

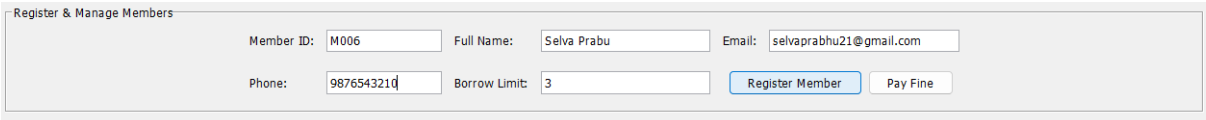


Figure 4: Search Book

The Search Book module allows users to search for a book using its Book ID or title. If the requested book exists, the application displays complete information including the title, author, category, and availability status.

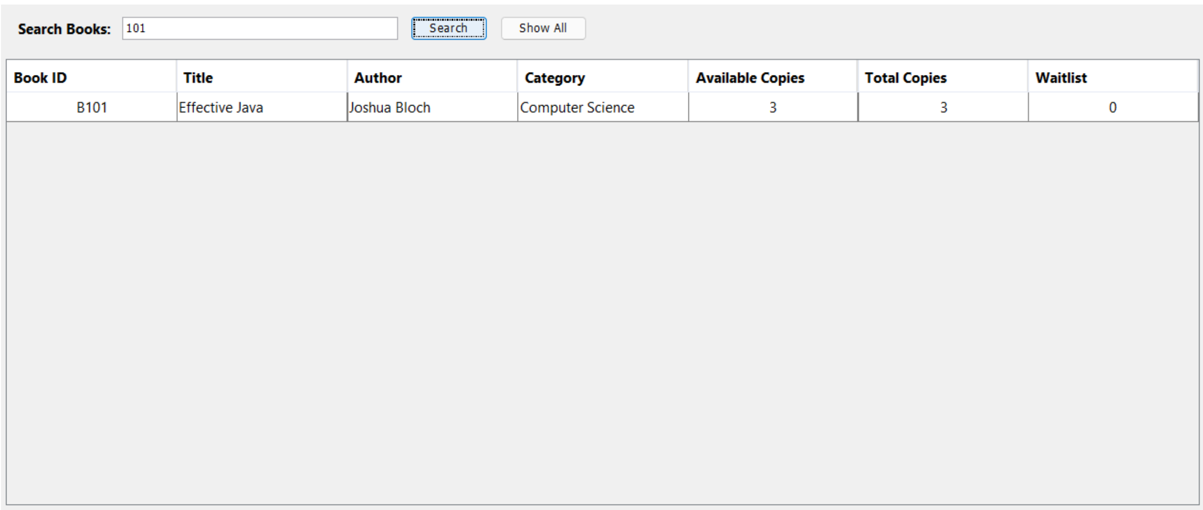


Figure 5: Issue Book

The Issue Book module allows the administrator to issue an available book to a registered member. The system verifies the book and member details, updates the inventory, records the transaction, and displays a confirmation message after successful issuance.

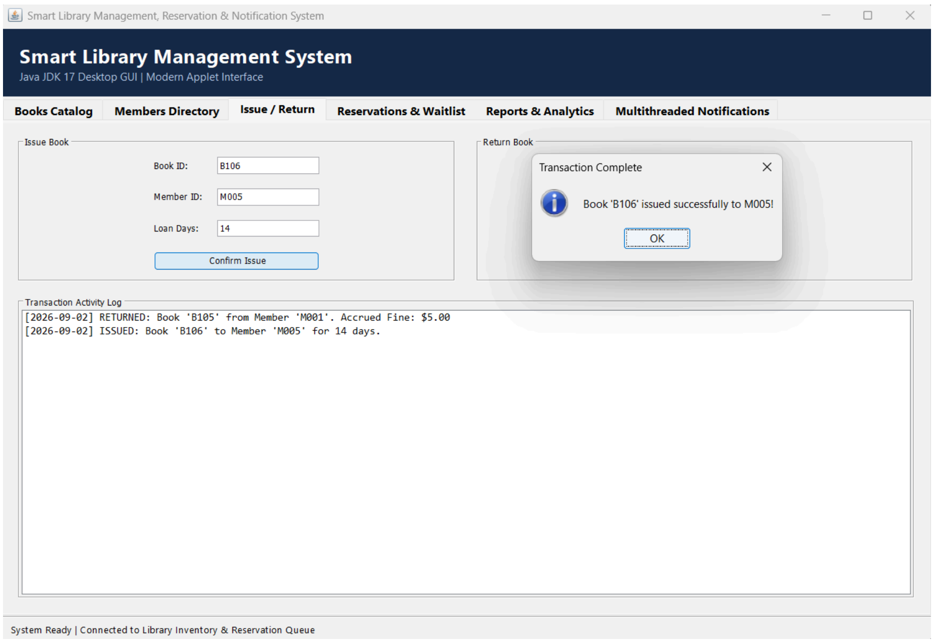


Figure 6: Return Book

The Return Book module processes the return of borrowed books. The application updates the inventory, checks whether the book was returned after the due date, calculates any applicable fine, and confirms the successful completion of the return process.

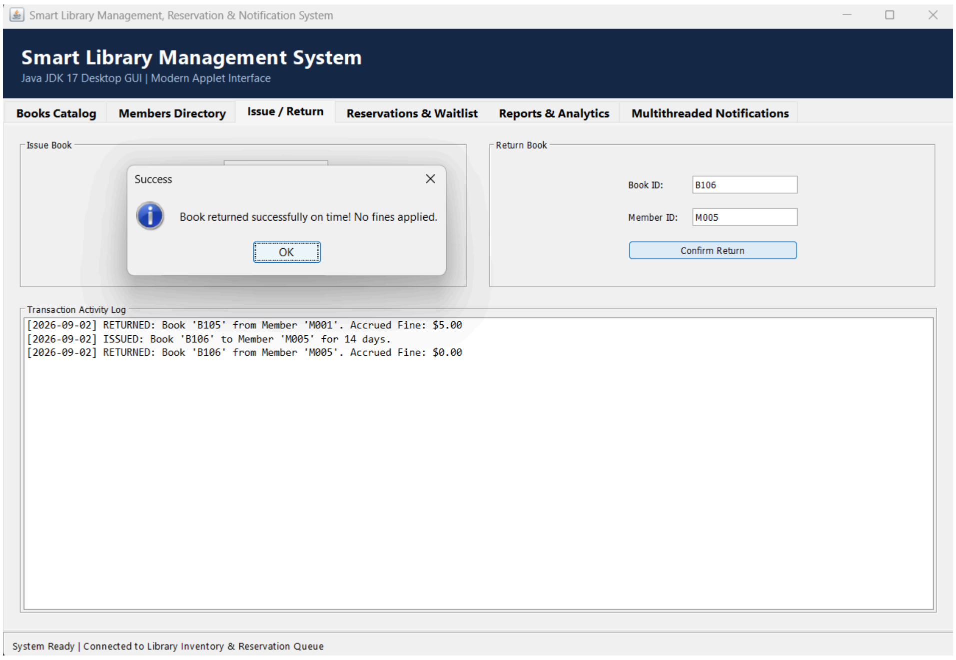


Figure 7: Reserve Book

The Reserve Book module allows members to reserve books that are currently unavailable. The reservation request is added to a waiting list, and the system confirms the reservation for future allocation when the book becomes available.

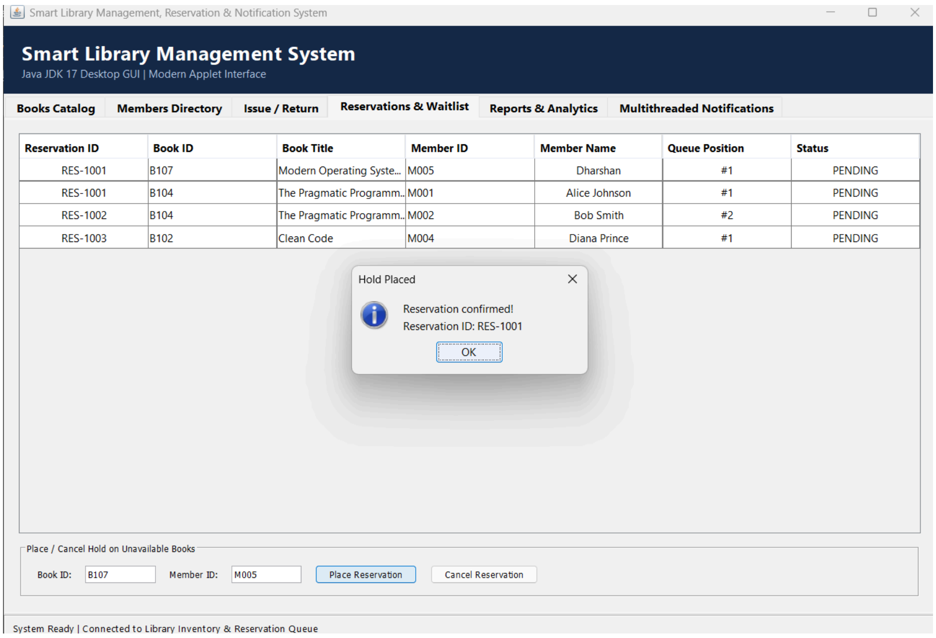


Figure 8: Cancel Reservation

The Cancel Reservation module enables members or the administrator to cancel an existing reservation. The system removes the reservation from the waiting list and updates the reservation records accordingly.

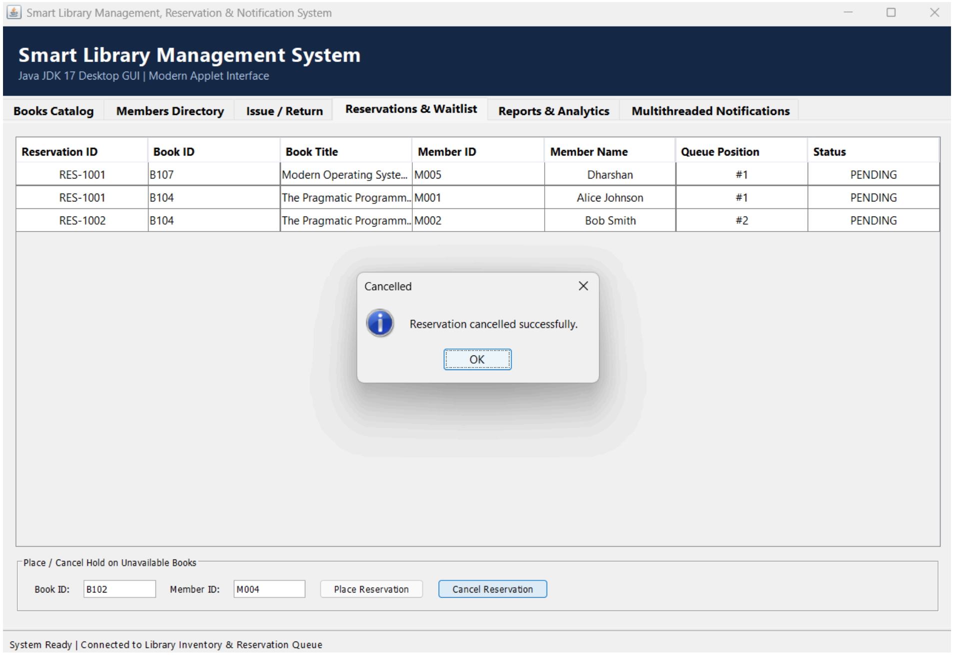


Figure 9: Display Books

The Display Books module shows the complete list of books available in the library. The displayed information includes Book ID, title, author, category, available copies, and current availability status, enabling efficient inventory management.

Book ID	Title	Author	Category	Available Copies	Total Copies	Waitlist
B109	Rich Dad Poor Son	Dharshan	Psycology	5	5	0
B107	Modern Operating Syste...	Andrew S. Tanenbaum	Operating Systems	4	4	1
B105	Artificial Intelligence	Stuart Russell	AI & Robotics	2	2	0
B1068	Human Antomy	Selva Prabhu	Fun	250	250	0
B106	Introduction to Algorithm...	Thomas H. Cormen	Algorithms	1	1	0
B103	Design Patterns	Erich Gamma et al.	Software Eng	2	2	0
B104	The Pragmatic Programm...	Andy Hunt	Software Eng	0	1	2
B101	Effective Java	Joshua Bloch	Computer Science	3	3	0
B102	Clean Code	Robert C. Martin	Computer Science	1	2	0

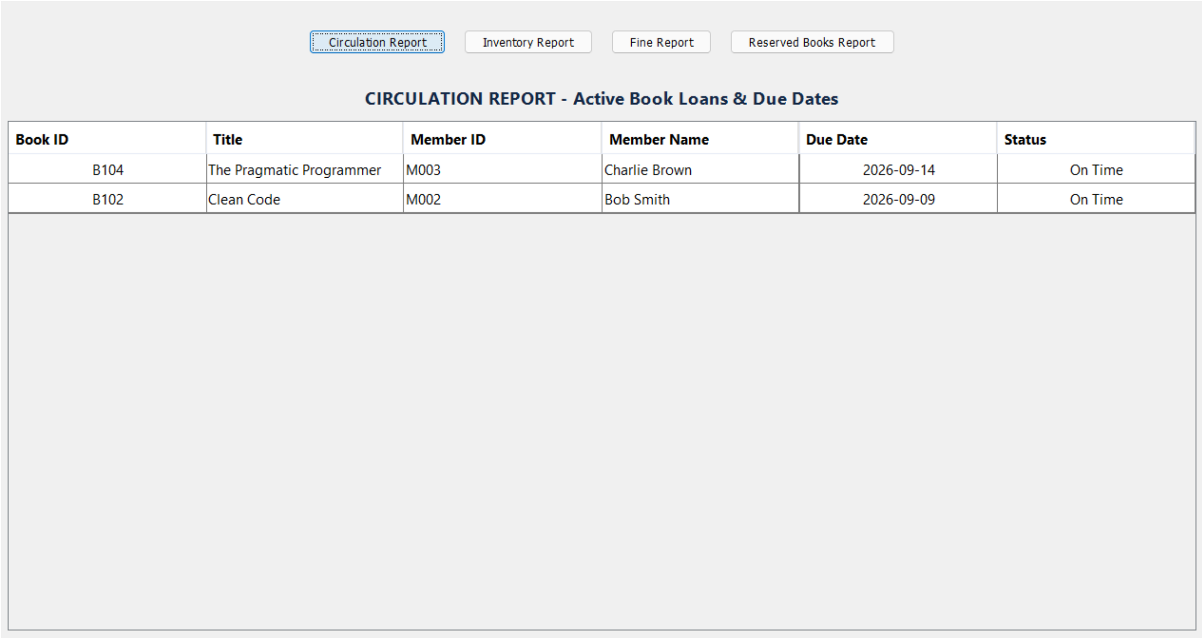
Figure 10: Display Members

The Display Members module lists all registered library members along with their Member ID, name, membership type, and other relevant information. This feature helps the administrator verify and manage member records efficiently.

Member ID	Name	Email	Phone	Borrowed / Limit	Outstanding Fine ($)
M004	Diana Prince	diana.p@university.edu	555-0104	0 / 4	$0.00
M003	Charlie Brown	charlie.b@university.edu	555-0103	1 / 2	$0.00
M002	Bob Smith	bob.smith@university.edu	555-0102	1 / 3	$0.00
M001	Alice Johnson	alice.j@university.edu	555-0101	0 / 3	$5.00
M006	Selva Prabu	selvaprabhu21@gmail.com	9876543210	0 / 3	$0.00
M005	Dharshan	dharshan2109@gmail.com	7305139670	0 / 3	$0.00

Figure 11: Generate Reports

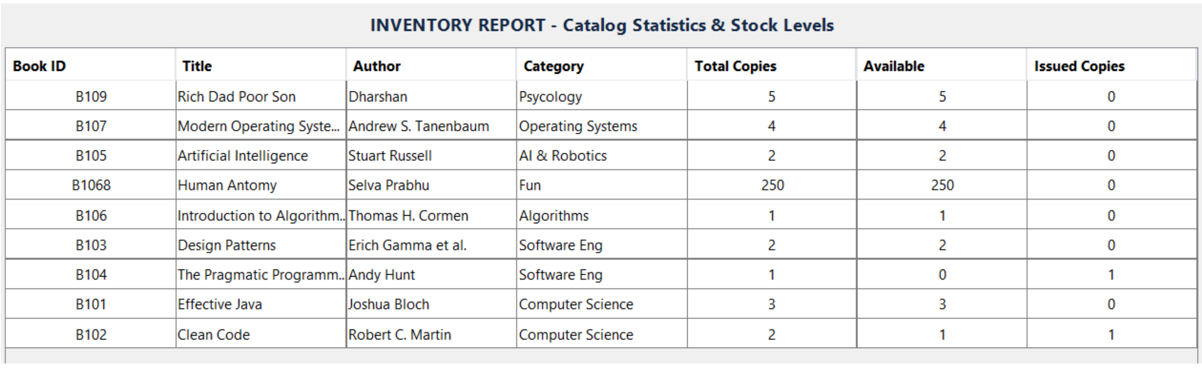
The Generate Reports module provides a summary of library activities, including issued books, returned books, available inventory, reservations, and overdue fines. These reports assist in monitoring library operations and evaluating resource utilization.



Circulation Report
Inventory Report
Fine Report
Reserved Books Report

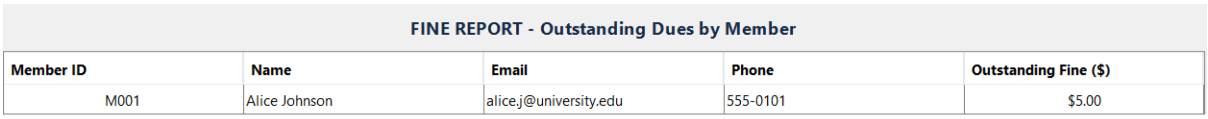
CIRCULATION REPORT - Active Book Loans & Due Dates

Book ID	Title	Member ID	Member Name	Due Date	Status
B104	The Pragmatic Programmer	M003	Charlie Brown	2026-09-14	On Time
B102	Clean Code	M002	Bob Smith	2026-09-09	On Time



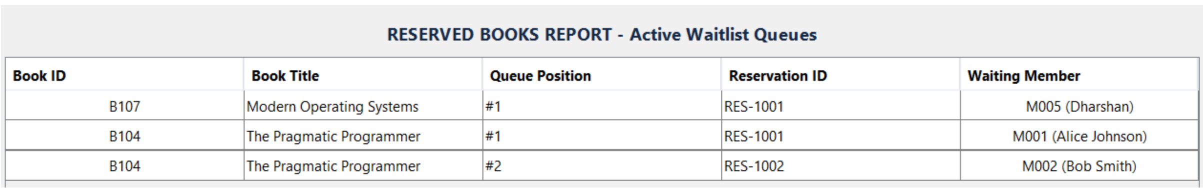
INVENTORY REPORT - Catalog Statistics & Stock Levels

Book ID	Title	Author	Category	Total Copies	Available	Issued Copies
B109	Rich Dad Poor Son	Dharshan	Psycology	5	5	0
B107	Modern Operating Syste...	Andrew S. Tanenbaum	Operating Systems	4	4	0
B105	Artificial Intelligence	Stuart Russell	AI & Robotics	2	2	0
B1068	Human Antomy	Selva Prabhu	Fun	250	250	0
B106	Introduction to Algorithm...	Thomas H. Cormen	Algorithms	1	1	0
B103	Design Patterns	Erich Gamma et al.	Software Eng	2	2	0
B104	The Pragmatic Programm...	Andy Hunt	Software Eng	1	0	1
B101	Effective Java	Joshua Bloch	Computer Science	3	3	0
B102	Clean Code	Robert C. Martin	Computer Science	2	1	1



FINE REPORT - Outstanding Dues by Member

Member ID	Name	Email	Phone	Outstanding Fine ($)
M001	Alice Johnson	alice.j@university.edu	555-0101	$5.00



RESERVED BOOKS REPORT - Active Waitlist Queues

Book ID	Book Title	Queue Position	Reservation ID	Waiting Member
B107	Modern Operating Systems	#1	RES-1001	M005 (Dharshan)
B104	The Pragmatic Programmer	#1	RES-1001	M001 (Alice Johnson)
B104	The Pragmatic Programmer	#2	RES-1002	M002 (Bob Smith)

Figure 12: Send Notifications

The Send Notifications module automatically generates due-date reminders and overdue alerts for members. Notifications are displayed through the console, helping members return books on time and improving overall library management.

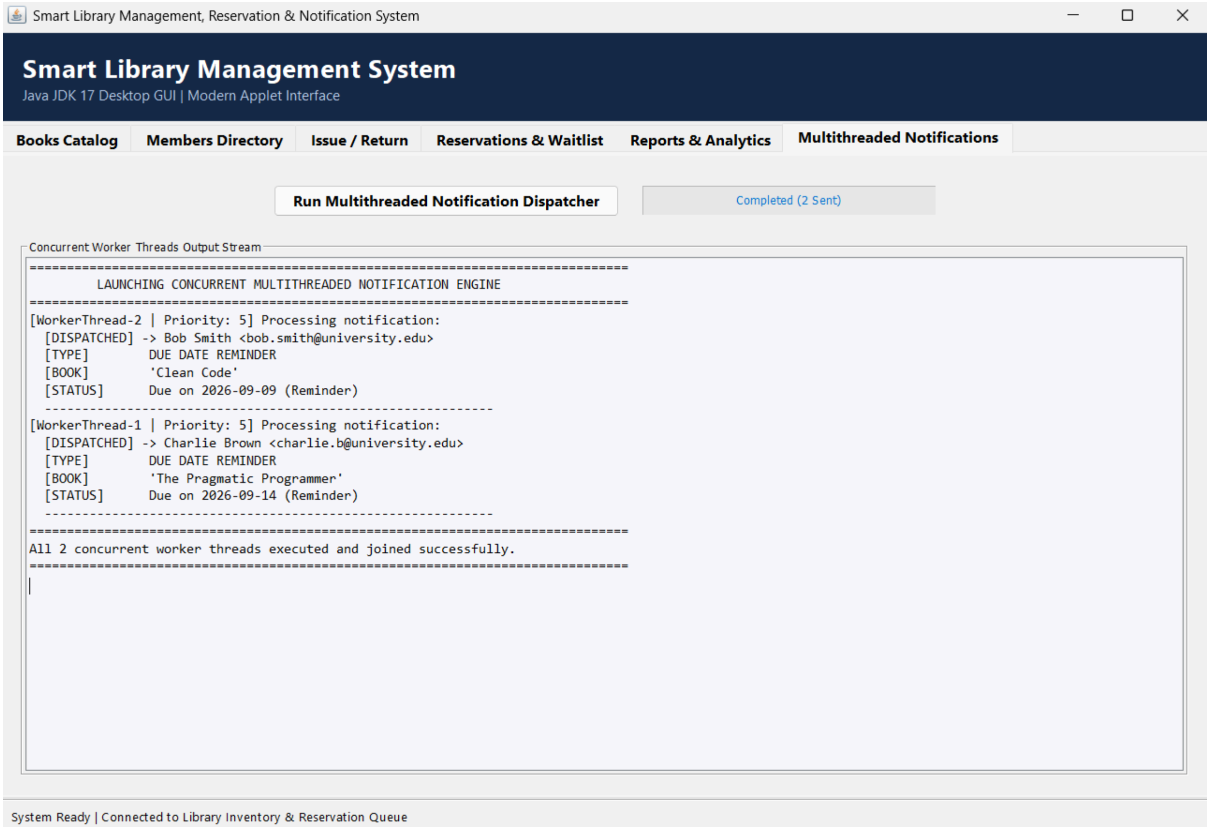
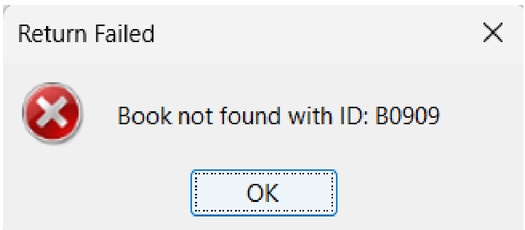


Figure 13: Exception Handling

The application validates user inputs before processing any operation. If an invalid Book ID, Member ID, duplicate reservation, or incorrect menu option is entered, the system displays an appropriate error message using Java exception handling mechanisms without terminating the application unexpectedly.



10. Results and Validation

Module	Result
Book Management	Successful
Member Management	Successful
Book Issue	Successful
Book Return	Successful
Reservation	Successful
Reports	Successful
Notifications	Successful

The application successfully performs all required operations. Testing confirmed that the implemented modules behave correctly, manage library records accurately, and generate the expected outputs, satisfying the assignment objectives.

11. Analysis, Comparison, Trade-offs and Justification

Advantages

- Fast execution
- Easy navigation
- Modular code
- Efficient searching
- Object-oriented design
- Supports concurrent notifications

Limitations

- GUI-based interface
- CSV files used as persistent database
- No database integration

Future Improvements

- Java Swing or JavaFX GUI
- MySQL database (future enhancement from current CSV storage)
- Email notifications
- Barcode scanning
- Online reservation portal

12. Broader Considerations / SDG Relevance

SDG 4 – Quality Education

Provides efficient access to library resources and supports learning.

SDG 9 – Industry, Innovation and Infrastructure

Uses modern software engineering concepts for digital library management.

SDG 16 – Peace, Justice and Strong Institutions

Maintains accurate records and transparent circulation management.

13. Conclusion

The Smart Library Management System successfully automates the daily operations of a library using Java. It demonstrates Object-Oriented Programming, Java Collections Framework, Exception Handling, and Multithreading while providing efficient book management, member registration, reservation handling, report generation, and notification services. The project satisfies the functional requirements of the assignment and provides a scalable foundation for future enhancements.

14. Individual Contribution of Group Members

Member	Contribution
Mohan Raj V D	Requirement analysis, project planning, GUI design, and documentation.
Monesh S	Developed the Book Management module (Add, Update, Delete, Search Books) and implemented CSV file handling for book records.
Selva Prabhu A K	Developed the Member Management module, Book Issue & Return module, and integrated CSV database operations for member and transaction records.
Ganesh	Implemented the Reservation, Notification, and Report Generation modules. Assisted with exception handling and GUI event handling.
Dharshan A N	Performed testing, debugging, validation, prepared screenshots, final report documentation, and coordinated project integration using Antigravity AI assistance.

15. References

1. Oracle Java SE Documentation
2. Java JDK 17 Documentation
3. Herbert Schildt, *Java: The Complete Reference*
4. Joshua Bloch, *Effective Java*
5. IntelliJ IDEA Documentation
6. Antigravity AI Documentation

16. One-Page Individual Reflection

This assignment helped me understand how Java programming concepts can be applied to develop a practical library management system. My main contribution was developing the Book Management module, which includes adding, updating, deleting, and searching book records. I also implemented CSV file handling to store and manage book-related information. Through this work, I gained practical experience in Object-Oriented Programming, file handling, data management, and designing modular functionalities. I also improved my understanding of exception handling and learned how different modules can work together as part of a complete application. Testing and debugging the Book Management functionalities helped me identify and resolve errors and improved my problem-solving skills. Working as part of a team also helped me understand the importance of coordination, modular development, and proper documentation. If given additional time, I would improve the module by integrating it with a proper database, enhancing the graphical user interface, and adding more advanced search and filtering features to make book management more efficient and user-friendly.